

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains the correct mean	AO1.1b	B1	1.38
(i)				
(ii)	Uses the correct formula for standard deviation	AO1.1a	M1	$\sqrt{\frac{261.8}{120} - 1.38^2}$
	Obtains the correct standard deviation	AO1.1b	A1	0.526 to 0.529
(b)	Uses the model to calculate a normal probability	AO3.4	M1	0.5417 to 0.5428
(i)	Obtains correct probability	AO1.1b	A1	
(ii)	Recalls correct value of 0	AO1.2	B1	0
(c)	Calculates the value of mean – 3 × standard deviations	AO1.1b	M1	–0.1998 to –0.207
	Concludes that model might be inappropriate as the value is less than 0	AO2.2b	A1	This is less than 0 and so model might not be appropriate
(d)	Standardises appropriately and formalises a probability statement PI by fully correct equation	AO3.1b	M1	$P\left(Z > \frac{0.75 - \mu}{0.21}\right) = 0.1$ $z \text{ value} = 1.2816$ $\frac{0.75 - \mu}{0.21} = 1.2816$ $\mu = 0.481$
	Obtains z value from inverse normal distribution (± 1.2816)	AO1.1a	M1	
	Forms a correct equation using standardised result and z value	AO1.1b	A1	
	Solves the equation to find the correct value of μ	AO1.1b	A1	
				Total 12 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	Finds mean	AO1.1b	B1	Mean $\bar{X} = \frac{3046.14}{100} = 30.46(14)$
	Finds standard deviation awfw (4.17 – 4.20) Allow $s = \sqrt{\frac{1746.29}{100}} = 4.179$	AO1.1b	B1	Standard deviation $s = \sqrt{\frac{1746.29}{99}} = 4.20$
	States max range for a normal distribution	AO1.2	M1	$\bar{X} + 3s \approx 43$ $\bar{X} - 3s \approx 18$
(ii)	Explains why normal model acceptable	AO2.4	E1	Normal dist model acceptable because data can be regarded as continuous and the range of amounts is within $\bar{X} \pm 3s$
	Finds probability awfw (0.095 – 0.097)	AO1.1b	B1	$P(X < 25) = 0.0967$
	Finds z from calculator	AO1.1b	B1	$P\left(Z < \frac{30 - 29.55}{\sigma}\right) = 0.55$
(b)	Uses Normal model to form equation	AO3.4	M1	$\frac{30 - 29.55}{\sigma} = 0.1257$
	Finds correct sd Correct accuracy required for this mark Accept 3.5 or 3.7	AO1.1b	A1	$\sigma = \frac{30 - 29.55}{0.1257} = 3.6$
				Total 8 marks

Q3.

	Marking Instructions	AO	Marks	Typical Solution
(a) (i)	States both hypotheses using correct language	AO2.5	B1	$H_0 : \mu = 123$ $H_1 : \mu \neq 123$
	Finds test statistic	AO1.1a	M1	$\frac{127 - 123}{70}$ Test statistic = $\sqrt{12144}$
	Obtains correct test statistic	AO1.1b	A1	= 6.30
	Infers H_0 rejected by comparison of ts with cv	AO2.2b	A1	Critical z values ± 1.96 $6.30 > 1.96$

	Concludes correctly in context, including 'some evidence'	AO3.2a	E1	Reject H_0 there is evidence (at the 5% level) to suggest the mean expenditure on bread had changed from 2012 to 2013
ALT	States both hypotheses using correct language	AO2.5	B1	$H_0 : \mu = 123$ $H_1 : \mu \neq 123$
	Attempts to find p value for z -test	AO1.1a	M1	From calculator, $P(X < 127) = 3.045 \times 10^{-10}$ $3.045 \times 10^{-10} < 0.025$
	Finds correct p value	AO1.1b	A1	
	Infers H_0 rejected by comparison of p with 0.025	AO2.2b	A1	
	Concludes correctly in context, including 'some evidence'	AO3.2a	E1	Reject H_0 there is evidence (at the 5% level) to suggest the mean expenditure on bread had changed from 2012 to 2013
(ii)	Uses Normal model to find critical values PI	AO3.4	M1	$123 \pm 1.96 \times \frac{70}{\sqrt{12144}}$
	Obtains correct critical values Correct accuracy required for this mark Disallow integer answers	AO1.1b	A1	min = 121.75 and max = 124.25
(b) (i)	States valid reason for statement 1 'not supported'	AO2.4	R1	The conclusion implies that the mean changed, not that it increased by a specific amount, so the statement is not supported
	Infers that model/test used would not imply the statement	AO2.2b	R1	
(ii)	States valid reason for statement 2 'not supported'	AO2.4	R1	The conclusion implies that there is evidence that the mean has changed, but expenditure increase may be due to price changes, so statement is not supported
	Infers that model/test used would not imply the statement	AO2.2b	R1	

Q4.

- (a) (i) Volume, $V \sim N(5.028, 0.015^2)$

In (a)(i), ignore the inclusion of a lower limit of 0; it has no effect on the answer

$$P(V < 5.04) = P\left(Z < \frac{5.04 - 5.028}{0.015}\right)$$

Standardising 5.04 with 5.028 and 0.015; allow (5.028 - 5.04)

M1

$$= P(Z < \underline{0.8})$$

*CAO; ignore inequality and sign
May be implied by a **correct** answer*

A1

$$= \underline{0.788}$$

AWRT (0.78814)

A1

(3)

- (ii) $P(V > 5) = P(Z > -1.87) = P(Z < +1.87)$

***Correct** area change
May be implied by a **correct** answer
or by an answer > 0.5*

M1

$$= \underline{0.968 \text{ to } 0.97}$$

AWFW (0.96903)

A1

(2)

5

- (b) 95% (0.95) $\Rightarrow z = \underline{1.96 \text{ or } 1.64 \text{ to } 1.65}$

AWRT / AFW; ignore sign

B1

$$z = \frac{(5.028 + v) - 5.028}{0.015}$$

*Standardising (5.028 $\pm$ v) or v
with 5.028 and 0.015*

M1

$$\frac{\pm v}{0.015} = \pm(1.96 \text{ or } 1.64 \text{ to } 1.65)$$

Allow inconsistent signs
Allow $\pm(v - 5.028)$

m1

$v = \underline{0.029 \text{ to } 0.03}$

AWFW
Must use consistent signs and 1.96
1.64 to 1.65 $\Rightarrow$ 0.024 to 0.025

A1

4

[9]

Q5.

- (a) Volume, $V \sim N(106, 2.5^2)$

In (a), ignore the inclusion of a lower limit of 0; it has no effect on the answer

$$P(V < 110) = P\left(Z < \frac{110 - 106}{2.5}\right)$$

Standardising 110 with 106 and 2.5;
allow (106 - 110)

M1

$$= P(Z < \underline{1.6})$$

CAO; ignore inequality and sign
*May be implied by a **correct** answer*

A1

$$= \underline{0.945}$$

AWRT (0.94520)

A1

3

- (b) $P(V > 100) = P(Z > -2.4) = P(Z < +2.4)$

Correct area change
*May be implied by a **correct** answer*
or by an answer > 0.5

M1

$$= \underline{0.991 \text{ to } 0.992}$$

AWFW (0.99180)

A1

2

(c) $P(104 < V < 108) = P(-a < Z < a) =$

$P(Z < a) - (1 - P(Z < a))$

or

$2 \times P(Z < a) - 1$

OE; $a = 0.8$ is **not** a requirement
 May be implied by **0.788** seen
 or by a **correct** answer

M1

$= 0.788 - (1 - 0.788) = 0.788 - 0.212$

or

$= 2 \times 0.788 - 1$

AWRT (0.78814 / 0.21186)
 Condone 0.211
 May be implied by a **correct** answer

A1

$= \underline{0.576}$

AWRT (0.57628)

A1

3

(d) $P(V \neq 106) = \underline{1 \text{ or one or unity or } 100\%}$

CAO; accept nothing else but ignore
 additional words providing they are
 not contradictory (eg certain so = 1)

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B1

1

[9]

Q6.

(a) $\sigma \approx \frac{10}{a} \text{ or } \frac{20}{b} \text{ or } \frac{\text{range}}{b} \text{ or } 10c \text{ or } 20d$

OE; with $2 \leq a \leq 4$ $4 \leq b \leq 8$
 or with c or d in equiv percentages
Cannot be implied from a correct
 answer (justification required)

M1

$\underline{2.5 \text{ or } 3.3(\text{OE}) \text{ or } 5}$

A1

SC Award B1 for only **2.5 or 3.3(OE)** or 5 with no justification
Award B0 for any other answer with no justification or with incorrect justification
(eg $\sqrt{10} = 3.16$)

(b) **Valid statement** involving:

391 and 405

OR

401 and 415

OR

24 and 10

OR

391 and 415 and 10 / 24 with linking statement

Allow 'set weight' to imply 415

and / or 'mean' to imply 391

B0 for 10 linked to σ

B1

95.5 > (value of σ of **2.5 or 3.3(OE) or 5**)

Accept $\neq$ rather than >

Clear correct numerical comparison

B1

Neither (likely to be) **correct**

Dependent on B1 B1

Bdep1

3

(c) Mean or $\bar{y} = \frac{8210.0}{10} = \underline{821}$
CAO;

OR

$\Sigma y = \underline{8200}$

B1

Variance $\frac{110.00}{9} = \underline{12.2}$
AWRT

or

$\frac{110.00}{10} = \underline{11}$

CAO

*Award on **value**; ignore notation*

OR

SD 3.5 or 3.3
 AWRT

B1

821 is similar to / within 10 of 820
 OE; **clear correct numerical**
 comparison of **821** with **820**
 Allow 'set weight' to imply 820

OR

Or

8210 is within 100 of 8200
 OE; **clear correct numerical**
 comparison of **8210** with **8200** but
 do **not** accept 'within 10' here

B1

3.5 or 3.3 is
 similar to
 a value of σ of 3.3(OE) or 2.5
 Clear correct numerical comparison

B1

4

[9]

Q7.

- (a) (i) Volume, $X \sim N(507.5, 4.0^2)$

$$P(X < 515) = P\left(Z < \frac{515 - 507.5}{4.0}\right)$$

Standardising 515 with 507.5 and 4.0 but allow (507.5 - 515)

M1

$$P(Z < 1.875) = \underline{\mathbf{0.97}}$$

AWRT

(0.96960)

A1

- (ii) $P(500 < X < 515) = P(-a < Z < a) =$

$$P(Z < a) - (1 - P(Z < a))$$

or

$$2 \times P(Z < a) - 1$$

OE; $a = 1.875$, $p = 0.97$ (AWRT) or correct standardising are not required May be implied by a **correct** answer

M1

$$= [(i) - (1 - (i)) \text{ or } (2 \times (i)) - 1] = \underline{\mathbf{0.94}}$$

AWRT (0.93921)

A1

(iii) $P(X \neq 507.5) = \underline{\mathbf{1 \text{ or one or unity or } 100\%}}$

CAO; accept nothing else but ignore additional words providing that they are not contradictory (eg certain so = 1)

B1

5

(b) $0.96 \Rightarrow z = \underline{\mathbf{1.75}}$

AWRT (ignore sign)(1.7507)

B1

$$\left(\frac{x - 507.5}{4.0} \right) = 1.5 \text{ to } 2.1$$

Standardising x with 507.5 and 4.0 but allow $(507.5 - x)$; **and** equating to a **z-value** (ignore signs)

M1

$x = \underline{\mathbf{514 \text{ to } 515}}$

AWFW (514.50)

Must be consistent signs throughout

A1

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3

(c) $p = \underline{\mathbf{(a)(ii)}} [= 0.94]$

F on (a)(ii) providing $0 < p < 1$

BF1

$$P(\geq 5 \text{ in } 6) = (6 \times p^5 \times (1 - p)) + p^6$$

Either expression with value of p

M1

$= \underline{\mathbf{0.948 \text{ to } 0.956}}$

AWFW (0.95296)

A1

3

Q8.

- (a) (i) Weight, $X \sim N(421, 2.5^2)$

Accept percentage equivalents in (a)

$$P(X = 421) = \underline{\mathbf{0 \text{ or zero or nought or } 0\%}}$$

CAO; accept nothing else but ignore additional words providing that they are not contradictory (eg impossible so = 0)

B1

$$(ii) \quad P(X < 425) = P\left(Z < \frac{425 - 421}{2.5}\right)$$

Standardising 425 with 421 and 2.5 but allow (421 - 425)

M1

$$= P(Z < 1.6) = \underline{\mathbf{0.945 \text{ to } 0.946}}$$

AWRT (0.94520)

A1

$$(iii) \quad P(418 < X < 424) = P(-a < Z < a) =$$

$$P(Z < a) - (1 - P(Z < a))$$

or

$$2 \times P(Z < a) - 1$$

OE; $a = 1.2$ or correct standardising are **not** required

May be implied by

0.885 (AWRT) seen anywhere

or by a correct answer

M1

$$= 0.885 - (1 - 0.885) = 0.885 - 0.115$$

$$\text{AWRT} \quad (0.88493 / 0.11507)$$

$$\text{or} = 2 \times 0.885 - 1$$

Implied by a **correct** answer

A1

$$= \underline{\mathbf{0.769 \text{ to } 0.77}}$$

$$\text{AWFW} \quad (0.76986)$$

A1

(a) Weight, $X \sim N(2200, 160^2)$

$$(i) \quad P(X < 2500) = P\left(Z < \frac{2500 - 2200}{160}\right)$$

Standardising 2500 with 2200 and 160; allow (2200 – 2500)

M1

$$= P(Z < 1.87 \text{ to } 1.88)$$

AWFW; ignore inequality and sign

May be implied by a correct answer

A1

$$= 0.969 \text{ to } 0.97(0)$$

AWFW

(0.96960)

A1

3

$$(ii) \quad P(X > 2000) = P(Z > -1.25)$$

$$= P(Z < +1.25)$$

Area change

May be implied by a correct answer or an answer > 0.5

M1

$$= 0.894 \text{ to } 0.895$$

AWFW

(0.89435)

A1

2

$$(iii) \quad P(2000 < X < 2500) = (i) - (1 - (ii))$$

OE; allow new start ignoring (i) & (ii)

Allow even if incorrect standardising providing $0 < \text{answer} < 1$

May be implied by a correct answer

M1

$$= 0.96960 - (1 - 0.89435) = 0.863 \text{ to } 0.865$$

AWFW

(0.86395)

A1

Note:

If (ii) is 0.105 to 0.106, then

$$(0.96960 - 0.10565) = 0.86395 \Rightarrow \text{M0 A0}$$

2

(b) $10\% \Rightarrow 90\% (0.90) \Rightarrow z = 1.28$

AWRT

(1.2816)

B1

$$z = \frac{1000 - 1125}{\sigma}$$

Standardising 1000 with 1125 and σ ; allow (1125 – 1000)

M1

$$= -1.2816$$

Only allow: ± 1.28

± 1.64 to ± 1.65

± 2.32 to ± 2.33

A1

$$\sigma = 97 \text{ to } 98$$

AWFW (97.53433)

Note:

Inconsistent signs $\Rightarrow$ B1 M1 A1 A0 max

A1

4

[11]

Q10.

$$151.5 \text{ cm} = 1.515 \text{ m} \approx 1.5 \text{ m}$$

OE

B1

Thus (could be) correct

OE; dependent on B1

Bdep1

$$n \times 26.6 > 10$$

OE; accept any $n \geq 1$

B1

Thus (likely to be) incorrect

OE; dependent on B1

Bdep1

[4]

Q11.

- (a) (i) Volume, $X \sim N(32, 10^2)$

$$P(X < 40) = P\left(Z < \frac{40-32}{10}\right)$$

Standardising 40 with 32 and 10; allow (32 - 40)

M1

$$= P(Z < 0.8)$$

CAO; ignore inequality and sign

May be implied by a correct answer

A1

$$= 0.788$$

AWRT (0.78814)

A1

3

- (ii) $P(X > 25) = P(Z < -0.7) = P(Z < +0.7)$

Area change

May be implied by a correct answer or an answer > 0.5

M1

$$= 0.758$$

AWRT (0.75804)

A1

2

- (iii) $P(25 < X < 40) = (i) - (1 - (ii))$

OE; allow new start ignoring (i) & (ii)

Allow even if incorrect standardising providing $0 < \text{answer} < 1$

*May be implied by a **correct** answer*

M1

$$= 0.78814 - (1 - 0.75804) = 0.546$$

Note:

If (ii) is 0.242, then $(0.788 - 0.242) = 0.546 \Rightarrow \text{M0 A0}$

AWRT (0.54618)

A1

2

- (b) $P(B < £65) = P\left(Z > \frac{48.5-32}{10}\right)$

Attempt to change from B to X using (48 to 49), 32 and 10

$$\text{or } P\left(Z > \frac{65 - 42.88}{13.4}\right)$$

*Attempt to work with distribution of B
 using 65, (42.8 to 42.9) and 13.4*

M1

$$= P(Z < 1.65) = 1 - P(Z < 1.65)$$

Area change

May be implied by a correct answer or an answer < 0.5

m1

$$= 1 - 0.95053 = 0.049 \text{ to } 0.05(0)$$

AWFW (0.04947)

A1

3

(c) Other **fuels**

Other **vehicles** with an example (not other cars)

Other types of **customer**

Minimum purchase (policy)

Purchases in integer / fixed £s

Customers filling fuel cans

Size of car / engine / fuel tank $\Rightarrow$ B0

Price of fuel $\Rightarrow$ B0

Customer paying capacity $\Rightarrow$ B0

*Must be **two clearly different** valid reasons for award of B2*

Drivers and vehicles related $\Rightarrow$ B1 eg lorry drivers & lorries

B2,1

2

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[12]

Q12.

***In (a)(i) & (c), ignore the inclusion of a lower
 limit of 0; it has no effect on either answer***

(a) Weight, $W \sim N(2.75, 0.15^2)$

$$(i) \quad P(W < 2.8) = P\left(Z < \frac{2.8 - 2.75}{0.15}\right)$$

Standardising 2.8 with 2.75 and

0.15; allow (2.75 - 2.8)

M1

$$= P(Z < \underline{\underline{0.33 \text{ or } 1/3}})$$

AWRT / CAO; ignore inequality and sign

*May be implied by a **correct** answer*

A1

$$= \frac{\mathbf{0.629 \text{ to } 0.633}}{\text{AWFW}} \quad (0.63056)$$

A1

(ii) $P(W > 2.5) = P(Z > -1.67) = P(Z < +1.67)$

Correct area change

*May be implied by a **correct** answer*

or an answer > 0.5

M1

$$= \frac{\mathbf{0.951 \text{ to } 0.953}}{\text{AWFW}} \quad (0.95221)$$

A1

5

(b) Weight, $X \sim N(5.25, 0.20^2)$

(i) $P(5.1 < X < 5.3) = P(Z < 0.25) - P(Z < -0.75)$

Must have diff of 2 probs for each B1

$$= \mathbf{0.59871}$$

Accept 0.599

B1

$$\mathbf{MINUS [(1 - 0.77337) \text{ or } 0.22663] = 0.372(08)}$$

Accept 0.773 or 0.227

*AG; do **not** mark simply on answer*

B1

2

(ii) $P(0 \text{ in } 4) = [1 - 0.372]^4$

Accept $[1 - c's (b)(i)]^4$

M1

$$= 0.628^4 = \mathbf{0.155 \text{ to } 0.156}$$

AWFW (0.15554)

A1

2

[9]

Q13.

Weight, $W \sim N(1018, 10^2)$

(a) $P(W < 1025) = P\left(Z < \frac{1025 - 1018}{10}\right)$
Standardising (1024.5, 1025 or 1025.5) with 1018
and ($\sqrt{10}$, 10 or 10^2) and / or ($1018 - x$)
May be gained in (a) or (b)

M1

$P(Z < 0.7) = 0.758$
AWRT (0.75804)

A1

2

(b) $P(1015 < W < 1030)$
 $= P(W < 1030) - P(W < 1015)$
 $= P(Z < 1.2) - P(Z < -0.3)$
Difference of two probabilities
May be implied

M1

$= 0.88493 - (1 - 0.61791)$
Area change
May be implied by answer

M1

$= 0.502 \text{ to } 0.504$
AWFW (0.50284)

A1

3

[5]

Q14.

- (a) Time taken is dependent upon leaving time
Or equivalent

B1

1

- (b) b (gradient) = 1.28 (or 141/110)
AWRT; (CAO or equivalent) (1.28182)

B2

b (gradient) = 1.25 to 1.35

AWFW

Treat rounding of correct answers as ISW

(B1)

a (intercept) = 29.95 to 30 (or 659/22)

AWFW; (CAO or equivalent) (29.95455)

B2

a (intercept) = 29 to 31

AWFW

(B1)

Thus $y = 30 + 1.28x$

F on a and b

B1F

5

or

Attempt at $\sum x$ $\sum x^2$ $\sum y$ $\sum y^2$ and $\sum xy$ ($\sum y^2$)

275 9625 682 and 20575 (47494)

(All four attempted)

or

Attempt at S_{xx} S_{xy} and S_{yy}

2750 and 3525 (5210)

(Both attempted)

(M1)

Attempt at correct formula for b gradient

©2025 Exam Papers Practice. All rights reserved. (m1)

b (gradient) = 1.28 (or 141/110)

AWRT; (CAO or equivalent)

(A1)

a (intercept) = 29.95 to 30 (or 659/22)

AWFW; (CAO or equivalent)

(A1)

Thus $y = 30 + 1.28x$

F on a and b

(B1F)

Accept a and b interchanged only if identified correctly

by a clearly shown equation

*If a and b are not identified anywhere in the question, then: 1.25 to 1.35 $\Rightarrow$ B1
 29 to 30 $\Rightarrow$ B1*

(c) $7.45 \text{ am} \Rightarrow x = 15$

CAO; stated, used or implied

B1

$\Rightarrow y_{15} = 30 + 1.28 \times 15$

Use of $10 < x < 20$

M1

$= 47 \text{ to } 52$

AWFW (49.2)

A1

Time before 9.00 am $= 9.00 - (7.45 + \text{c's } y_{15})$

May be implied

M1

$= 23 \text{ to } 28$

AWFW (25.8)

A1

5

SC Answer of 17 CAO
 (use of c's $y_{15} = 58$) gains 2 marks

NB: *An answer of 8.32 to 8.37
 gains B1 M1 A1 M0 A0*

(d) (i) $y_{85} = 30 + 1.28 \times 85 = 135 \text{ to } 146$

AWFW (138.9)

B1

1

(ii) Extrapolation / outside/ above range of x -values

Or equivalent

B1

Implies leaves home at 8.55 so different traffic conditions

*Or equivalent; 8.55 may be implied by
 5 minutes*

B1

2

[14]

Q15.

- (a) Volume, $V \sim N(412, 8^2)$

(i) $P(V < 400) = P\left(Z < \frac{400 - 412}{8}\right)$
Standardising 400 with 412 and 8 and / or
(412 - x)

M1

$= P(Z < -1.5) = 1 - P(Z < 1.5)$
Area change
May be implied by a correct answer or an
answer < 0.5

M1

$= 1 - 0.93319 = 0.066 \text{ to } 0.067$
AWFW (0.06681)

A1

3

(ii) $P(V > 420) = P(Z > 1)$
CAO but ignore inequality and sign
May be implied by a correct answer

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B1

$= 1 - P(Z < 1) = 1 - 0.84134$
 $= 0.158 \text{ to } 0.159$
AWFW (0.15866)

B1

2

(iii) $P(V = 410) = 0$ or zero or impossible
Ignore any working
B0 for 'impossible to calculate' or 'no answer'

B1

1

- (b) (i) A statement / indication that
 (–) 1.6449 and / or 2.3263 are z-values
Simple statement that 1.6449 $z = \pm$ and / or
 $z = \pm 2.3263$
or sketch of normal curve

with at least one z-value marked

B1

Do **not** allow $\Phi(0.99) = 2.3263$, etc but **allow**
 $\Phi^{-1}(0.99) = 2.3263$

Do **not** award for z-value(s) simply
 embedded in standardisation statement(s)

A clear use of $z = \frac{v - \mu}{\sigma}$ or $\mu + 2\sigma$ with 400 and/or

420 (condone sign errors)

SC Immediate algebraic use of
 $v - \mu = z\sigma \Rightarrow$ B1 M1 A0

M1

The two given equations correctly derived
 AG; watch for sign inconsistencies

A1

3

- (ii) Thus $20 = (2.3263 + 1.6449)\sigma$
*A sensible (one that would lead to values required
 if completed correctly) attempt at solving the
 two given equations by eliminating μ or σ*
 Do **NOT** allow MC or MR

M1

$$\sigma = 5.04$$

AWRT (5.03626)

A1

$$\mu = 408$$

AWRT (408.284)

A1

3

[12]

Q16.

- (a) Weight, $W \sim N(165, 2.5^2)$

(i)
$$P(W < 167) = P\left(Z < \frac{167 - 165}{2.5}\right)$$

Standardising 167 with 165 and 2.5; allow (165 – 167)

M1

$$= P(Z < 0.8)$$

*CAO; ignore inequality and sign
May be implied by a correct answer*

A1

$$= 0.788$$

AWRT (0.78814)

A1

3

$$(ii) \quad P(W > 162) = P(Z > -1.2) \\ = P(Z < 1.2)$$

*Area change; may be implied by correct answer or
answer > 0.5*

M1

$$= 0.884 \text{ to } 0.886$$

AWFW (0.88493)

A1

2

$$(b) \quad P(12 \text{ pucks} < 167) = p^{12} \quad \text{with} \quad 0 < p < 1$$

*Any probability to power 12 or $1 - p^{12}$;
do **not** allow multiplying factors*

M1

$$= [(a)(i)]^{12} = (0.78814)^{12} = 0.057 \text{ to } 0.058$$

AWFW (0.05744)

A1

2

$$(c) \quad 1\% \Rightarrow 99.5\% (0.995) \Rightarrow z = 2.57 \text{ to } 2.58$$

AWFW (2.5758)

B1

$$z = \frac{170 - 165}{\sigma} \quad \text{or} \quad \frac{160 - 165}{\sigma}$$

Standardising 170 or 160 with 165 and σ ; allow (165 – x)

M1

$$= 2.5758 \text{ or } -2.5758$$

Only allow: ± 2.05 to ± 2.06
 ± 2.32 to ± 2.33
 ± 2.57 to ± 2.58

A1

$\sigma = 1.94$

AWRT (1.94114)

A1

Note:

Inconsistent signs $\Rightarrow$ B1 M1 A1 max

4

[11]

Q17.

- (a) (i) Diameter, $D \sim N(57.15, 0.04^2)$

$$P(D < 57.2) = P\left(Z < \frac{57.2 - 57.15}{0.04}\right)$$

Standardising 57.2 with 57.15 and 0.04; allow (57.15 – 57.2)

M1

$$= P(Z < 1.25)$$

*CAO; ignore inequality and sign
May be implied by a **correct** answer*

A1

$$= 0.894 \text{ to } 0.895$$

AWFW (0.89435)

A1

3

- (ii) $P(57.1 < D < 57.2)$

$$= p - (1 - p)$$

*Allow even if incorrect standardising providing
 $p - (1 - p)$ **seen**
May be implied by a correct answer*

M1

$$= 2 \times 0.89435 - 1 = 0.788 \text{ to } 0.79(0)$$

AWFW (0.78870)

A1

2

- (b) $P(16 \text{ balls} < 57.2) = p^{16}$ with $0 < p < 1$
*Any probability to power 16 or $1 - p^{16}$;
 do **not** allow multiplying factors
 If only seen in (b)(ii), allow just M1*

M1

$$= [(a)(i)]^{16} = (0.89435)^{16} = \mathbf{0.166 \text{ to } 0.17(0)}$$

AWWF (0.16754)

A1

2

[7]

Q18.

- (a) (i) $X \sim N(10.2, 0.15^2)$

$$P(X < 10.5) = P\left(Z < \frac{10.5 - 10.2}{0.15}\right)$$

*Standardising (10.45, 10.5 or 10.55) with 10.2
 and ($\sqrt{0.15}$, 0.15 or 0.15^2) and/or $(10.2 - x)$*

M1

$$= P(Z < 2)$$

*CAO; ignore inequality and sign
 May be implied by a correct answer*

A1

$$= 0.977$$

AWRT (0.97725)

A1

3

- (ii) $P(10.0 < X < 10.5)$
 $= [C's (a)(i)] - P(X < 10.0)$

*Or equivalent; must be clear correct
 method if answer incorrect and
 answer > 0*

M1

$$= (a)(i) - P(Z < -1.33)$$

$$= (a)(i) - (1 - p)$$

*Method correct using -1.3
 gives 0.88 to 0.881*

$\Rightarrow$ M1 m1 A0

$$= 0.97725 - (1 - 0.90824)$$

Area change

May be implied by a correct answer or answer > 0.5

m1

$$= 0.885 \text{ to } 0.887$$

AWFW (0.88604)

A1

M1 m1 A1 for $0.90824 - [1 - (a)(i)] = 0.886$

M1 m0 A0 for $(a)(i) - 0.90824 = 0.0685$

M0 m0 A0 for answer < 0

3

(b) $P(X > 10) = p[\text{from (a)(ii)}]$

$$= 0.908 \text{ to } 0.909$$

Correct value or F on value used or implied in (a)(ii) providing > 0.5

Use of - 1.3 gives 0.9032

B1F

$$P(6 \text{ rolls} > 10) = 0.90824^6$$

Accept any probability to power 6

M1

$$0.56 \text{ to } 0.565$$

AWFW

A1

Note:

B0F M1 A0 is possible

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3

[9]

Q19.

Volume, $X \sim N(153, 1.6^2)$

(a) $P(\text{not overflow}) = P(X < 155)$

May be implied

B1

$$= P\left(Z < \frac{155 - 153}{1.6}\right)$$

Standardising (154.5, 155 or 155.5 or 149.5, 150 or 150.5) with 153 and

$(\sqrt{1.6}, 1.6 \text{ or } 1.6^2) \text{ and/or } (153 - x)$
May be gained in (a)(i) or (a)(ii)

M1

$$= P(Z < 1.25)$$

CAO; ignore inequality and sign
May be implied by a correct answer

A1

$$= 0.894 \text{ to } 0.895$$

AWFW (0.89435)

A1

(b) $P(\text{less than printed}) = P(X < 150)$

Only if B1 not awarded in (a)(i)

(B1)

$$= P(Z < -1.875)$$

$$1 - P(Z < 1.875)$$

Area change
May be implied by a correct answer
or answer < 0.5

M1

$$= 0.03 \text{ to } 0.031$$

AWFW (0.03040)

A1

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[6]

Q20.

Time. $X \sim N(65, 20^2)$

(a) (i) $P(X < 90) = P\left(Z < \frac{90 - 65}{20}\right) =$
 $\left[P\left(Z < \frac{0 - 65}{20}\right) = P(Z < -3.25) = 0.00058 \right]$

Standardising (89.5, 90 or 90.5 or 59.5,
60 or 60.5) with 65 and $(\sqrt{20}, 20 \text{ or } 20^2) \text{ and/or } (65 - x)$
May be gained in (a)(i) or (a)(ii)

M1

$$= P(Z < 1.25)$$

*CAO; ignore inequality and sign
 May be implied by a correct answer*

A1

$$= 0.893 \text{ to } 0.895$$

AWFW (0.89435)

A1

(ii) $P(X > 60) = P(Z > -0.25)$

$$= P(Z < 0.25)$$

*Area change
 May be implied by a correct answer
 or answer > 0.5*

M1

$$= 0.598 \text{ to } 0.599$$

AWFW (0.59871)

A1

5

(b) (i) $P(1 \text{ in } 6 = 60) = \mathbf{0 \text{ or zero or impossible}}$

*Ignore any working
 B0 for 'impossible to calculate'*

B1

1

(ii) $P(X < 60) = 1 - [(a)(ii)] \text{ or } (0.401 \text{ to } 0.402)$

May be implied

M1

$$P(6 \text{ in } 6 < 60) = p^6 \text{ with } 0 < p < 1$$

*Any probability to power 6; do **not**
 allow multiplying factors*

M1

$$= (0.40129)^6 = \mathbf{0.004 \text{ to } 0.00425}$$

Dependent on M1 M1 (0.0041759)

A1dep

3

[9]

Q21.

(a) Indication that 2.0537 is

{a / the 98% (0.98)} z-value

B1

Standardising 45 using μ and σ : $z = \frac{45 - \mu}{\sigma}$ **or** $45 = \mu + z\sigma$
Or equivalent

M1

Fully correct derivation (including any necessary re-arrangement to given expression)

A1

3

- (b) 0.95 (95%) $\Rightarrow z = 1.64$ to 1.65
AWFW; ignore sign (1.6449)

B1

Hence $30 - \mu = -1.6449\sigma$

Subtracting two equations
Or equivalent

M1

$15 = 3.6986\sigma$
 Thus $\sigma = 4(.0)$ to 4.1
AWFW (4.0556)

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A1

Thus $\mu = 36.6$ to 36.7
AWFW (36.6710)

A1

4

[7]

Q22.

$X \sim N(5.08, 0.05^2)$

- (a) (i) $P(X < 5) = P\left(Z < \frac{5 - 5.08}{0.05}\right) = P(Z < -1.6)$
Standardising (4.5, 4.95, 5, 5.05 or 5.5)
with 5.08 and ($\sqrt{0.05}$, 0.05 or 0.05^2)
and/or ($5.08 - x$)

M1

$$= 1 - P(Z < 1.6) = 1 - 0.9422$$

*Area change; may be implied
AWFW (0.0548)*

m1

$$= 0.0545 \text{ to } 0.055$$

$(1 - \text{answer}) \Rightarrow \text{M1 max}$

A1

3

(ii) $P(5 < X < 5.10) = P(X < 5.10) - (i)$

*Or equivalent; must be clear correct
method if answer incorrect and answer > 0*

M1

$$= P(Z < 0.4) - (i) = 0.65542 - 0.0548$$

$$= 0.6 \text{ to } 0.601$$

AWFW (0.60062)

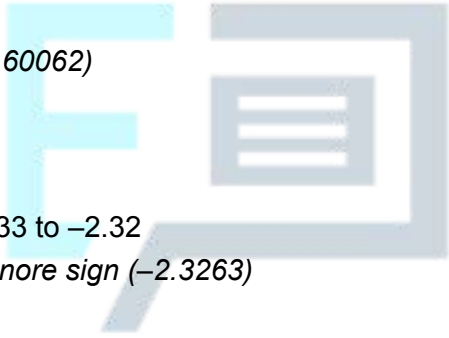
A1

2

(b) $1\% (0.01) \Rightarrow z = -2.33 \text{ to } -2.32$

AWFW; ignore sign (-2.3263)

B1


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$$z = \frac{5 - \mu}{0.05}$$

$$= -2.3263$$

*Standardising 5 with μ and 0.05 or
0.025; allow $(\mu - 5)$*

M1

*Only allow: ± 2.05 to ± 2.06
 ± 2.32 to ± 2.33
 ± 2.57 to ± 2.58*

A1

$$\mu = 5.11 \text{ to } 5.12$$

AWFW (5.1163)

A1

Note:

$$\frac{5 - \mu}{0.05} = 2.3263 \Rightarrow 5.116$$

$\Rightarrow$ B1 M1 A1 A0

Or equivalent inconsistent signs

4

[9]

Q23.

(a) $W \sim N(3.12, 0.08^2)$

$$P(2.95 < X < 3.20) = P\left(\frac{2.95 - 3.12}{0.08} < Z < \frac{3.20 - 3.12}{0.08}\right)$$

Standardising (2.945, 2.95 or 2.955) or (3.195, 3.20 or 3.205)

with 3.12 and ($\sqrt{0.08}$, 0.08 or 0.08^2) and/or $(3.12 - x)$

M1

$$= P(-2.125 < Z < 1)$$

Either; CAO 1
AWFW -2.13 to -2.12

A1

$$= P(Z < 1) - [1 - P(Z < 2.125)]$$

Area change; may be implied

m1

$$= 0.84134 - [1 - (0.98300 \text{ to } 0.98341)]$$

$$= 0.824 \text{ to } 0.825$$

AWFW (0.82455)

(1 - answer) $\Rightarrow$ M1 A1 max

A1

4

(b) 2.5% (0.975) $\Rightarrow z = -1.96$

AWRT; ignore sign (-1.9600)

B1

$$z = \frac{3 - 3.15}{\sigma}$$

Standardising 3 with 3.12
and σ ; allow $(3.12 - 3)$

M1

$$= -1.96$$

Only allow: ± 1.96
 ± 1.64 to ± 1.65

A1

$$\sigma = 0.06 \text{ to } 0.0613$$

AWFW (0.06122)

A1

Note:

$$\frac{3 - 3.12}{\sigma} = 1.96 \Rightarrow \sigma = 0.06122$$

$\Rightarrow$ B1 M1 A1 A0

Or equivalent inconsistent signs

4

[8]

Q24.

(a) $X \sim N(253, 5^2)$

(i)
$$P(X < 250) = P\left(Z < \frac{250 - 253}{5}\right) =$$

Standardising (249.5, 250 or
 250.5) with 253 and ($\sqrt{5}$, 5 or 5^2)
 and/or (253 - x)

M1

$$P(Z < -0.6) = 1 - P(Z < 0.6) = 1 - 0.72575$$

Area change; may be implied

m1

$$= 0.274 \text{ to } 0.275$$

AWFW (0.27425)

(1 - answer) $\Rightarrow$ M1 max

A1

3

(ii) $P(245 < X < 250) = [C's(a)(i)] - P(X < 245)$

Or equivalent; must be clear
 correct method if answer incorrect
 and answer > 0

M1

$$= (i) - P(Z < -1.6) = 0.27425 - 0.0548$$

$$= 0.219 \text{ to } 0.22(0)$$

AWFW (0.21945)

M1 A0 for $[1 - (i)] - 0.0548 = 0.67095$

M0 A0 for $0.9452 - [(i)] = 0.67095$

M1 A1 for $0.9452 - [1 - (i)] = 0.21945$

A1

2

(iii) $P(X = 245) = 0$ or zero or impossible

Ignore any working

B0 for 'for impossible to calculate'

B1

1

(b) 98% (0.98) $\Rightarrow z = -2.05$ to -2.06

AWFW; ignore sign (-2.0537)

B1

$$z = \frac{245 - 253}{\sigma}$$

Standardising 245 with 253 and σ ; allow (253 - 245)

M1

$$= -2.0537$$

Only allow: ± 2.05 to ± 2.06
 ± 2.32 to ± 2.33

A1

$$\sigma = 3.88 \text{ to } 3.9(0)$$

AWFW (3.8954)

A1

Note:

$$\frac{245 - 253}{\sigma} = 2.0537 \Rightarrow \sigma = 3.8954$$

$\Rightarrow$ B1 M1 A1 A0

Or equivalent inconsistent signs

4

[10]

Q25.

(a) $P(X < 3.5) = P\left(Z < \frac{3.5 - 3.3}{0.16}\right) =$

*Standardising (3.45, 3.5 or 3.55) with 3.3
 & ($\sqrt{0.16}$, 0.16 or 0.16^2) and/or $(3.3 - x)$*

M1

$P(Z < 1.25) =$

CAO; ignore sign

A1

0.849 to 0.895

AWFW (0.89435)

A1

3

(b) $P(X > 3.0) = P\left(Z > \frac{3.0 - 3.3}{0.16}\right) =$

*Standardising (2.95, 3 or 3.05) with 3.3
 & ($\sqrt{0.16}$, 0.16 or 0.16^2) and/or $(3.3 - x)$*

M1

$P(Z > -1.875) = P(Z < 1.875) =$

Correct area change

m1

0.969 to 0.97(0)

AWFW (0.96960)

A1

3

(c) $P(3.0 < X < 3.5) = (i) - [1 - (ii)] =$
OE

M1

0.863 to 0.865

AWFW; CSO (0.86395)

A1

2

[8]

Q26.

(a) (i) $P(X < 3.5) = P\left(Z < \frac{3.5 - 3.3}{0.16}\right) =$
Standardising (3.45, 3.5 or 3.55) with 3.3
& ($\sqrt{0.16}$, 0.16 or 0.16^2) and/or $(3.3 - x)$

M1

$P(Z < 1.25) =$
CAO; ignore sign

A1

0.894 to 0.895
AWFW (0.89435)

A1

3

(ii) $P(X > 3.0) = P\left(Z > \frac{3.0 - 3.3}{0.16}\right) =$
Standardising (2.95, 3 or 3.05) with 3.3
& ($\sqrt{0.16}$, 0.16 or 0.16^2) and/or $(3.3 - x)$

M1

$P(Z > -1.875) = P(Z < 1.875) =$
Correct area change

m1

0.969 to 0.97(0)
AWFW (0.96960)

A1

3

(iii) $P(3.0 < X < 3.5) = (i) - [1 - (ii)] =$
OE

M1

0.863 to 0.865
AWFW: CSO (0.86395)

A1

2

(b) $0.025 \Rightarrow z = 1.96$
CAO; ignore sign

B1

$$z = \frac{3.1 - \mu}{0.16}$$

*Standardising 3.1 with μ and 0.16;
allow $(\mu - 3.1)$*

M1

$$= -1.96$$

*Equating z-term to z-value;
not using 0.025, 0.975, $|1 - z|$
or $\Phi(0.025) = 0.507$ to 0.512*

m1

$$\text{Hence } \mu = 3.4(0) \text{ to } 3.42$$

AWFW; CSO (3.4136)

A1

4

[12]

Q27.

Length $L \sim N(69.5, 0.55^2)$

$$(a) \quad (i) \quad P(L < 70) = P\left(Z < \frac{70 - 69.5}{0.55}\right) =$$

Standardising (69.5, 70 or 70.5) with

*69.5 and $(\sqrt{0.55}, 0.55 \text{ or } 0.55^2)$ and/or rights reserved.
(69.5 - x)*

M1

$$P(Z < 0.91) =$$

0.91 AWRT; ignore sign

A1

$$0.818 \text{ to } 0.82(0)$$

AWFW (0.81835)

A1

3

$$(ii) \quad P(69 < L < 70) =$$

$$P(L < 70) - P(L < 69) =$$

$$P(Z < 0.91) - P(Z < -0.91) =$$

Difference (70 – 69)

M1

$$P(Z < 0.91) - \{1 - P(Z < 0.91)\} =$$

$$(0.81835) - (1 - 0.81835) =$$

Correct area change

m1

$$0.636 \text{ to } 0.64(0)$$

AWFW (0.63670)

A1

3

(iii) $P(L = 70) = 0$

CAO

B1

1

(b) $0.90 \text{ (90\%)} \Rightarrow z = -1.28$

AWRT; ignore sign (-1.2816)

B1

$$z = \frac{l - 69.5}{0.55}$$

Standardising l with 69.5 and 0.55;
allow (69.5 – l)

M1

$$= \pm 1.28(16)$$

Equating z -term to the z -value

A1

Hence $l = 68.7 \text{ to } 68.9$

AWFW; CSO (68.796)

A1

4

(c) Variance of $\bar{L}_{20} = \frac{0.55^2}{20} = 0.0151(25)$

$$\bar{L}_{20} = \frac{0.55}{\sqrt{20}} = 0.123$$

SD of

CAO/AWRT; stated or used

B1

$$P(\bar{Z}_{20} > 69.25) = P\left(Z > \frac{69.25 - 69.5}{\sqrt{0.55^2/20}}\right)$$

Standardising 69.25 with 69.5 and 0.123;
allow (69.5 – 69.25)

M1

$$= P(Z > -2.03) = P(Z < 2.03) =$$

Correct area change

m1

0.978 to 0.98(0)
AWFW (0.97896)

A1

4

[15]

Q28.

Height $X \sim N(140, 2.5^2)$

$$(a) \quad (i) \quad P(X < 145) = P\left(Z < \frac{145 - 140}{2.5}\right) =$$

Standardising (144.5, 145 or 145.5) with
140 and ($\sqrt{2.5}$, 2.5 or 2.5²) and/or (140 – x)

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$$P(Z < 2) =$$

2 CAO; ignore sign

A1

0.977 to 0.98(0)
AWFW (0.97725)

A1

3

$$(ii) \quad P(138 < X < 142) = P(X < 142) - P(X < 138) =$$

Difference (142 – 138)

M1

$$P(Z < 0.8) - P(Z < -0.8) =$$

0.8 CAO

B1

$$P(Z < 0.8) - \{1 - P(Z < 0.8)\} =$$

$$(0.78814) - (1 - 0.78814) =$$

Correct area change

m1

$$0.576 \text{ to } 0.58(0)$$

AWFW (0.57628)

A1

4

(b) $0.85 \text{ (85\%)} \Rightarrow z = -1.03 \text{ to } -1.04$

AWFW; ignore sign (-1.0364)

B1

$$z = \frac{x - 140}{2.5}$$

Standardising x with 140 and 2.5;
allow $(140 - x)$

M1

$$= \pm 1.03 \text{ to } \pm 1.04$$

Equating z -term to the z -value

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A1

$$\text{Hence } x = 137.3 \text{ to } 137.5$$

AWFW; CSO (137.41)

A1

4

[11]

Q29.

(a) (i) $P(X < 45) = P\left(Z < \frac{45 - 37}{8}\right)$

Standardising (44.5, 45 or 45.5) with 37
and $(\sqrt{8}, 8 \text{ or } 8^2)$ and/or $(37 - x)$

M1

$$= P(Z < 1)$$

CAO; ignore sign

A1

$$= 0.841$$

AWRT (0.84134)

A1

3

(ii) $P(30 < X < 45) = (i) - P(X < 30)$

Used; OE

M1

$$= (i) - P(Z < -0.875)$$

$$= (i) - [1 - (0.80785 \text{ to } 0.81057)]$$

Area change

m1

$$= 0.648 \text{ to } 0.652$$

AWFW (0.65056)

A1

3

(b) $0.12 \Rightarrow z = 1.17 \text{ to } 1.18$

AWFW; ignore sign (1.1750)

B1

$$z = \frac{45 - 40}{\sigma}$$

Standardising 45 with 40 and σ

M1

$$= 1.175$$

*Equating z-term to z-value but not using
0.12, 0.88 or $|1 - z|$*

m1

$$\sigma = 4.23 \text{ to } 4.28$$

AWFW

A1

4

(c) **Route A:** $P(X > 45) = 1 - (a)(i)$

Route B: $P(Y > 45) = 0.12$
OE; must use 45

B1

so

↑Dep↑

Monica should use **Route B** (smaller prob)
ft on (a)(i); allow Route Y

B1ft

2

[12]

Q30.

(a) (i) $P(X < 45) = P\left(Z < \frac{45-37}{8}\right)$
Standardising (44.5, 45 or 45.5) with 37
and ($\sqrt{8}$, 8 or 8^2) and/or $(37 - x)$

M1

$= P(Z < 1)$
CAO; ignore sign

A1

$= 0.841$
AWRT (0.84134)

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A1

3

(ii) $P(30 < X < 45) = (i) - P(X < 30)$
Used; OE

M1

$= (i) - P(Z < -0.875)$

$= (i) - [1 - (0.80785 \text{ to } 0.81057)]$
Area change

m1

$= 0.648 \text{ to } 0.652$
AWFW (0.65056)

A1

3

- (b) $0.12 \Rightarrow z = 1.17 \text{ to } 1.18$
AWFW; ignore sign (1.1750)

B1

$$z = \frac{45 - 40}{\sigma}$$

Standardising 45 with 40 and σ

M1

$$= 1.175$$

*Equating z-term to z-value but not using
 0.12, 0.88 or $|1 - z|$*

m1

$$\sigma = 4.23 \text{ to } 4.28$$

AWFW

A1

4

- (c) **Route A:** $P(X > 45) = 1 - (a)(i)$
Route B: $P(Y > 45) = 0.12$
OE; must use 45

B1

so EXAM PAPERS PRACTICE

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Monica should use **Route B** (smaller prob)
ft on (a)(i); allow Route Y

B1ft

2

- (d) Mean of $\overline{W} = 18$
*CAO; can be implied by **use** in standardising*

B1

Variance of $\overline{W} = \frac{12^2}{36} = 4$

CAO; OE

B1

$$P(\bar{w} > 20) = P\left(Z > \frac{20 - 18}{2}\right)$$

Standardising 20 with 18 and 2 and/or (18 – 20)

M1

$$= P(Z > 1) = 0.159$$

AWRT (0.15866); ft on (a)(i) if used

A1ft

4

(e) In part (d)

CAO; OE

B1

1

[17]

Q31.

(a) Time, $X \sim N(48, 20^2)$

$$(i) \quad P(X < 60) = P\left(Z < \frac{60 - 48}{20}\right) =$$

*Standardising (59.5, 60 or 60.5) with 48 and
($\sqrt{20}$, 20 or 20^2) and/or (48 – x)*

M1

$$P(Z < 0.6) = 0.725 \text{ to } 0.73$$

AWFW (0.72575)

A1

2

$$(ii) \quad P(30 < X < 60) = \\ P(X < 60) - P(X < 30) =$$

$$(i) - P(X < 30) =$$

$$(i) - P(Z < -0.9) =$$

Difference or equivalent

Standardising other than 60 and 30

$\Rightarrow$ max of M1 m1 A0

M1

$$(i) - \{1 - P(Z < +0.9)\} =$$

$$0.72575 - \{1 - 0.81594\} =$$

Area change

m1

0.54 to 0.542

AWFW (0.54169)

A1

3

(iii) $0.9 \Rightarrow z = 1.28$ to 1.282

AWFW (1.2816)

B1

$$z = \frac{k - 48}{20}$$

Standardising k with 48 and 20

M1

= 1.2816

*Equating z-term to z-value; not using 0.9,
0.1, $|1 - z|$ or $\Phi(0.9) = 0.81594$*

m1

$k = 73.6$ to 74

AWFW

A1

4

(b) Time, $Y \sim N(37, 25^2)$

(i) Use of $\mu - (2 \text{ or } 3) \times \sigma =$
 $37 - (50 \text{ or } 75)$

Or equivalent justification

M1

$< 0 \Rightarrow$ likely negative time

for (likely) negative times

B1

2

(ii) Central Limit Theorem or n large / > 30

B1

1

- (iii) Variance of $\bar{Y} = \frac{25^2}{35}$
OE; stated or used

B1

$$P(\bar{Y} > 40) = P\left(Z > \frac{40 - 37}{25/\sqrt{35}}\right) =$$

*Standardising 40 with 37 and 25/ $\sqrt{35}$
 and/or (37 – 40)*

M1

$$P(Z > 0.71) = 1 - P(Z < 0.71) =$$

Area change

m1

0.238 to 0.24
AWFW (1 – 0.76115)

A1

4

[16]

Q32.

- (a) (i) Time, $X \sim N(55, 8^2)$

$$P(X < 60) = P\left(Z < \frac{60 - 55}{8}\right)$$

*Standardising (59.5, 60 or 60.5) with 55 and
 ($\sqrt{8}$, 8 or 8²)
 and/or (55 – x)*

M1

$$= P(Z < 0.62 \text{ to } 0.63)$$

AWFW; ignore sign 0.625

A1

$$= 0.732 \text{ to } 0.736$$

AWFW 0.73401

A1

3

- (ii) $P(55 < x < 60) = (i) - P(x < 55) = (i) - 0.5$

Difference OE

M1

$$= 0.232 \text{ to } 0.236$$

AWFW 0.23401

A1

2

(b) Time, $Y \sim N(\mu, 16^2)$

$$0.95 \Rightarrow z = 1.64 \text{ to } 1.65$$

AWFW 1.6449

B1

$$P(Y < 90) = P\left(Z < \frac{90 - \mu}{16}\right)$$

Standardising 90 using μ and 16

M1

Thus $\frac{90 - \mu}{16} = 1.6449$

*Equating z-term to z-value;
Not using 0.95, 0.05 or $|1 - z|$*

m1

Thus $\mu = 63.6 \text{ to } 63.8$

AWFW

A1

4

[9]

Q33.

(a) Weight, $X \sim N(406, 4.2^2)$

$$P(X < 400) =$$

$$(i) \quad P\left(Z < \frac{400 - 406}{4.2}\right)$$

*Standardising (399.5, 400 or 400.5) with
406 and ($\sqrt{4.2}$, 4.2 or 4.2²) and/or (406 - x)*

M1

$$P(Z < -1.428 \text{ to } -1.43)$$

$$= 1 - P(Z < 1.428 \text{ to } 1.43)$$

$$\Phi(-z) = 1 - \Phi(z)$$

m1

$$= 0.076 \text{ to } 0.077$$

$$\text{AWRT} \quad 0.07636$$

A1

3

(ii) $P(402.5 < X < 407.5) =$
 $P(X < 407.5) - P(X < 402.5) =$
Difference OE

M1

$$P(Z < 0.36) - P(Z < -0.83)$$

AWRT; ignoring signs

B2,1

$$= 0.64058 - (1 - 0.79673) = 0.433 \text{ to } 0.44$$

$$\text{AWFW} \quad 0.43731$$

A1

4

(b) (i) $0.975 \Rightarrow z = 1.96$

Accept explanation in words

M1

$$P(Y < 310) = P\left(Z < \frac{310 - \mu}{\sigma}\right)$$

Standardising 310 using μ and σ

M1

or

$$X = \mu \pm z\sigma$$

Accept in words

$$\text{Thus } \frac{310 - \mu}{\sigma} = 1.96\sigma \Rightarrow \text{result}$$

Equating

m1

3

or

$$310 = \mu + 1.96\sigma \Rightarrow \text{result}$$

Substitution

NB: Working backwards from given equation $\Rightarrow$ at most M1 M0 m0

$$(ii) \quad 0.86 \Rightarrow z = 1.08$$

AWRT 1.0803

B1

$$310 - \mu = 1.96\sigma$$

$$307.5 - \mu = 1.08\sigma$$

$$2.5 = 0.88\sigma$$

Attempt at solving 2 equations each of form $x - \mu = z\sigma$

M1

$$\sigma = 2.84 \text{ to } 2.842$$

AWFW 2.841

A1

$$\mu = 304.4 \text{ to } 304.5$$

AWFW 304.43

A1

4

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[14]

Q34.

Height, $X \sim N(185, 10^2)$

$$(a) \quad P(X < 200) = P\left(Z < \frac{200 - 185}{10}\right)$$

standardising (199.5, 200 or 200.5) with 185

and ($\sqrt{10}$, 10 or 10^2) and/or $(185 - x)$

M1

$$= P(Z < 1.5)$$

CAO; ignore sign

A1

$$= \Phi(1.5) = 0.933$$

AWRT (0.93319)

A1

3

$$(b) \quad P(X > 175) = P\left(Z > \frac{175 - 185}{10}\right)$$

standardising (174.5, 175 or 175.5) with 185 and
($\sqrt{10}$, 10 or 10^2) and/or (185 - x)

M1

$$= P(Z > -1) = P(Z < 1)$$

area change

m1

$$= 0.841$$

AWRT (0.84134)

A1

3

$$(c) \quad P(175 < X < 200) = (i) - [1 - (ii)]$$

or equivalent

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M1

$$= 0.93319 - [1 - 0.84134]$$

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$$= 0.774 \text{ to } 0.775$$

AFWW (0.77453)
 ft on (i) and (ii) providing > 0

A1ft

2

[8]

Q35.

$$(a) \quad (i) \quad X \sim N(\mu, 4^2)$$

$$\mu = 106$$

$$P(X < 110) = P\left(Z < \frac{110 - 106}{4}\right)$$

Standardising (109.5, 110 or 110.5) with 106 and ($\sqrt{4}$, 4 or 4²) and/or (106 – x)

M1

$$= P(Z < 1)$$

CAO; ignore sign

A1

$$= 0.841$$

AWRT (0.84134)

A1

3

(ii) $P(\text{underweight}) = P(X < 100)$

Use of AFWW 99 to 100

M1

$$= P(Z < -1.5) = 1 - \Phi(1.5)$$

Area change

m1

$$= 1 - 0.93319 = 0.0668 \text{ to } 0.067$$

AWFW (0.06681)

A1

3

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(b) $2\% \Rightarrow z = -2.0537$

AWFW 2.05 to 2.06; ignore sign

B1

$$z = \frac{100 - \mu}{4}$$

M1

Thus $\frac{100 - \mu}{4} = -2.0537$

m1

Thus $\mu = 108.2 \text{ to } 108.3$

Standardising AFWW 99 to 100 with μ and 4

*Equating z-term to z-value;
not using 0.02, 0.98 or $|1 - z|$
AWFW*

A1

4

[10]



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